Example 10.14

Solve : $\frac{dy}{dx} = \sqrt{4x+2y-1}$.

Solution

By putting $z = 4x+2y-1$, we have

$$\frac{dz}{dx} = 4+2\frac{dy}{dx} = 4+2\sqrt{z}$$

$$\text{Hence} \quad \frac{dz}{4+2\sqrt{z}} = dx .$$

$$\text{Integrating,} \quad \int\frac{dz}{4+2\sqrt{z}} = x+C .$$

$$\text{Putting } z = u^2 \text{, we have}$$

$$\int\frac{dz}{4+2\sqrt{z}} = \int\frac{udu}{u+2} = u-2\log|u+2|+C,$$

$$\text{or} \quad \sqrt{z}-2\log\left|\sqrt{z}+2\right| = x+C$$

From which on substituting $z = 4x+2y-1$, we have the general solution

$$\sqrt{4x+2y-1}-2\log\left|\sqrt{4x+2y-1}+2\right| = x+C .$$

Example 10.15

Solve: $\frac{dy}{dx} = \frac{x-y+5}{2(x-y)+7}$.

Solution

Given that $\frac{dy}{dx} = \frac{x-y+5}{2(x-y)+7}$

Put $z = x-y$

$$\frac{dz}{dx} = 1-\frac{dy}{dx}$$

$$\frac{dy}{dx} = 1-\frac{dz}{dx}$$

Thus, the given equation reduces to

$$1-\frac{dz}{dx} = \frac{z+5}{2z+7}$$

$$\frac{dz}{dx} = 1-\frac{z+5}{2z+7}$$

$$\frac{dz}{dx} = \frac{z+2}{2z+7}$$

Separating the variables, we get

$$\frac{2z+7}{z+2}dz = dx$$

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